Freight Rate Prediction — Validation Report

Spotter ML Assessment · Two-part model (Ridge + LightGBM)

1. Objective

Predict posted_rate for 12,000 held-out loads (Nov–Dec 2025) and produce a fixed-lane December 2025 rate chart (Lexington → Fort Wayne, 360 miles, Dry Van, 32,000 lb).

2. Approach to validate train / test data and split

Labeled development data in train-test.csv covers 2025-01-01 through 2025-10-31 (48,000 loads). Unlabeled final loads in validation.csv cover 2025-11-01 through 2025-12-31. Because the real scoring window is future months, validation is time-forward — not a random shuffle (which would leak near-term market conditions).

Split	Dates	Rows	Use
Fit	Jan–Aug 2025	38,477	Model fitting / contamination stats
Holdout	Sep–Oct 2025	9,523	Model selection & holdout metrics
Final train	Jan–Oct 2025 (cleaned)	47,323	Refit before submission predictions

2.1 Data validation / cleaning

- weight: absolute value (sign errors), then median fill.
- market_index: same-date median fill (row vs date-mean correlation ≈ 0.99).
- RPM contamination filter (TRAIN ONLY): for each equipment × distance quantile bin, drop rows whose rate-per-mile is outside 0.5–1.8× the bin's typical median (~1.4% of rows; 677 of 48,000 on full Jan–Oct). Holdout/validation labels are never dropped for scoring — only training labels are cleaned.
- December chart features: city coords from training map; market_index by date from train+validation panels; Dry Van median quote_signal.

2.2 Model and holdout validation metrics

Both parts predict log(rate per mile). Final price = distance × exp(part1 + part2).

- Part 1 — Ridge: log distance, equipment, weight, market_index, months-since-start, quarter-end ramp (extrapolates market drift into Nov–Dec).
- Part 2 — LightGBM residual: route/load detail only — no calendar features (distance, weight, equipment, lat/lon, deltas, quote_signal).

Model	Target	Full MAE	MedAE	MAPE
LightGBM	log rate per mile	116.03	27.24	4.0%

Fixed December Prediction Chart (score.py)

3. Scorer command

python score.py --predictions validation_predictions.csv --december-predictions

data/december_chart_inputs.csv

The scorer validated 12,000 final predictions and 31 fixed December predictions, then wrote scorer_results/candidate_december.png (embedded below). Fixed inputs: Lexington → Fort Wayne | 360 miles | Dry Van | 32,000 lb | only date changes.

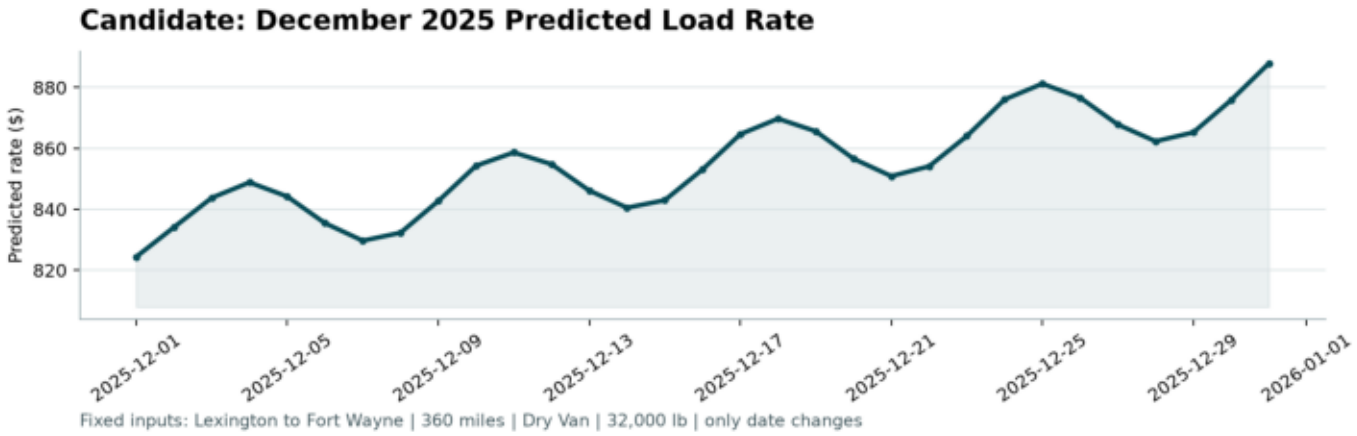


Figure 1. Candidate: December 2025 Predicted Load Rate
(produced by the provided score.py)

4. How to reproduce

python3 -m venv .venv

source .venv/bin/activate

python -m pip install -r requirements.txt

python train_predict.py --model two_part

python score.py --predictions validation_predictions.csv --december-predictions

data/december_chart_inputs.csv